

Large Language Models: An Applied Econometric Framework*

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Abstract

Large language models (LLMs) enable researchers to analyze text at unprecedented scale and minimal cost. Researchers can now revisit old questions and tackle novel ones with rich data. We provide an econometric framework for realizing this potential in two empirical uses. For prediction problems – forecasting outcomes from text – valid conclusions require “no training leakage” between the LLM’s training data and the researcher’s sample, which can be enforced through careful model choice and research design. For estimation problems – automating the measurement of economic concepts for downstream analysis – valid downstream inference requires combining LLM outputs with a small validation sample to deliver consistent and precise estimates. Absent a validation sample, researchers cannot assess possible errors in LLM outputs, and consequently seemingly innocuous choices (which model, which prompt) can produce dramatically different parameter estimates. When used appropriately, LLMs are powerful tools that can expand the frontier of empirical economics.

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1 Introduction

Large language models (LLMs) enable economists to process text at unprecedented scale and minimal cost, unlocking questions previously out of reach: predicting stock returns from earnings call transcripts, measuring partisan polarization in social media posts, backcasting historical consumer sentiment from newspapers, or simulating survey responses cheaply. By making these questions – and countless others – tractable, LLMs offer transformative potential for empirical research.

To realize this potential, we must answer a practical question: how should LLM outputs be incorporated into empirical workflows? Can they be plugged in “as-is,” or do our estimation strategies require adjustment to use these powerful tools?

This question is challenging because LLMs resist traditional statistical analysis. LLMs are complex, often proprietary, constantly evolving, and trained on sprawling, heterogeneous corpora that defy tractable modeling – extraordinary engineering achievements accomplished without our usual statistical foundations. Existing evaluation methods have been remarkably effective for developing better models but offer limited guidance for how any given LLM will perform on a new task.

We develop an econometric framework that addresses these complexities and provides practical guidance for empirical research using LLM outputs, focusing on two empirical uses: prediction problems and estimation problems. For each use, we clarify what assumptions and practices ensure valid conclusions.

In prediction problems, researchers use collected text to predict some economic outcome (e.g., predicting stock prices from corporate earnings calls). Extracting meaning from text requires modeling the complex structure of language. LLMs, having already learned this structure from enormous training corpora, can possibly serve as the foundation upon which economists can build in prediction problems — either by directly prompting an LLM to make a prediction or by using its representations as features in a predictor.

Using LLMs in prediction problems requires one condition: “no training leakage.” If a researcher forms predictions using an LLM and evaluates those predictions on an evaluation dataset, this reflects the LLM’s out-of-sample performance if and only if there is no overlap between the LLM’s training dataset and the researcher’s dataset. This can be violated because many LLMs are trained on intentionally obscured datasets. Computer scientists have found that LLMs are often trained on common benchmark evaluations. We find that LLMs appear to be trained on economic datasets; see also [Glasserman and Lin \(2023\)](#); [Sarkar and Vafa \(2024\)](#); [Lopez-Lira, Tang and Zhu \(2025\)](#) among others.

We provide guidance on enforcing no training leakage, clarifying that it requires careful

attention to both the choice of model and research design. For example, when the goal is to predict future, unseen documents (such as tomorrow’s financial news), open-source LLMs with fixed, published weights (e.g., [Touvron et al., 2023](#); [Dubey et al., 2024](#)) or time-stamped training data (e.g., [Sarkar, 2024](#); [He et al., 2025](#)) can be paired with evaluation samples constructed only from documents published after the model’s publication date or training cutoff.

In estimation problems, researchers estimate relationships between economic concepts expressed in text (e.g., partisanship in social media posts) and downstream parameters (e.g., the causal effect of a policy change). There is *some* resource-intensive procedure for measuring the economic concept; if it could be scaled, we would use these measurements and report the resulting estimates. Often, this would involve the researcher carefully reading each text piece themselves. We would like to instead use an LLM to economize on measurement costs. How can we use LLM outputs for valid downstream inference?

We highlight a constructive solution borrowing from an old idea in econometrics: collect a small validation sample and use it to *empirically* correct for possible LLM errors. We illustrate how researchers can incorporate validation data in the context of linear regression. Debiasing LLM outputs preserves our usual econometric guarantees of consistency and asymptotic normality. This has been well-studied in econometrics (e.g., [Lee and Sepanski, 1995](#); [Chen, Hong and Tamer, 2005](#); [Schennach, 2016](#)) and extended in machine learning, such as [Wang, McCormick and Leek \(2020\)](#), [Angelopoulos et al. \(2023\)](#), [Egami et al. \(2024\)](#), [Battaglia et al. \(2024\)](#), and [Carlson and Dell \(2025\)](#). We illustrate that incorporating debiased LLM outputs can substantially improve the precision of downstream estimates – resulting in estimates that are often more precise than using validation data alone. Used correctly, imperfect LLM outputs serve not as substitutes but as amplifiers of validation samples, allowing researchers to achieve tighter standard errors.

Absent a validation sample, researchers cannot assess the magnitude or pattern of errors in LLM outputs—and therefore cannot evaluate their impact on downstream parameter estimates. We demonstrate this problem empirically: absent a validation sample, seemingly innocuous choices—which LLM to use, how to phrase the prompt—lead to dramatically different parameter estimates in applications to finance and political economy, with coefficients varying in magnitude, sign, and significance. In estimation problems, researchers have no choice but to invest effort and collect validation samples when using LLMs.

Finally, this framework is flexible enough to account for creative uses of LLMs in economics. We argue that using LLMs for hypothesis generation (e.g., [Ludwig and Mullenbath, 2024](#)) can be cast as a form of prediction problem, so that the key assumption required is again no training leakage. LLM simulation of human subjects in surveys or ex-

periments (e.g., [Park et al., 2023](#); [Horton, 2023](#); [Manning, Zhu and Horton, 2024](#); [Park et al., 2024](#)) can be thought of as an estimation problem, and so in-silico subjects can amplify – but not fully replace – a small validation sample of actual human responses.

Our framework clarifies when and how researchers can harness LLMs in empirical research. The requirements are straightforward: ensure no training leakage for prediction and collect small validation samples for estimation. We provide a checklist based on our framework in [Section 6](#). These simple practices unlock the transformative potential of LLMs for empirical research.

2 An Econometric Framework for Large Language Models

LLMs are complex architectures trained on vast corpora through multi-stage processes: pre-training (learning to predict next tokens), instruction fine-tuning, reinforcement learning from human feedback (RLHF), and reinforcement learning from verifiable rewards (RLVR). Reasoning models employ test-time computation to further improve performance. The field evolves rapidly, with new architectures, training procedures, and capabilities emerging regularly.¹

This complexity creates a fundamental challenge for econometric analysis. We typically study statistical procedures by articulating assumptions about the data-generating process and combining them with a computational understanding of the procedure. This approach is at present intractable for LLMs. They have billions of parameters, proprietary training datasets, and algorithms that resist formal characterization.

We adopt a different strategy based on how economists actually use these tools. We treat LLMs as black boxes – prompting them or extracting embeddings without characterizing internal mechanisms – and identify conditions they must satisfy for valid empirical use. We focus on two applications: prediction problems and estimation problems. By black-boxing their inner workings, our approach provides guidance that is robust to inevitable changes in architectures and training procedures.

2.1 Setting and the Researcher’s Dataset

Let Σ^* denote a collection of strings (up to some finite length) in an alphabet with elements $\sigma \in \Sigma^*$. A training dataset is any collection of strings, summarized by the vector t whose elements t_σ are sampling indicators for whether string σ was collected.

¹Many resources are available about the technical foundations of large language models. See, for example, [Chang et al. \(2024\)](#), [Minaee et al. \(2025\)](#), and [Zhao et al. \(2025\)](#). For overviews aimed at economists, see [Korinek \(2023\)](#), [Dell \(2024\)](#), [Korinek \(2024\)](#), and [Ash, Hansen and Muvdi \(2024\)](#).

For any empirical question, only some strings are economically relevant. Denote these as $\mathcal{R} \subseteq \Sigma^*$ with elements $r \in \mathcal{R}$ that we refer to as text pieces. The researcher’s dataset is summarized by the vector d whose elements d_σ are sampling indicators for whether the researcher collected string σ . The researcher only collects economically relevant text pieces, and so $d_\sigma = 0$ for all $\sigma \in \Sigma^* \setminus \mathcal{R}$.

Each text piece r can be linked to observable economic variables (Y_r, W_r) , which are economic outcomes Y_r that might be influenced by the text or candidate covariates W_r that might influence the text.

Example: Congressional legislation Consider descriptions of bills introduced in the United States Congress. Each text piece $r \in \mathcal{R}$ is a bill’s description such as “A bill to revise the boundary of Crater Lake National Park in the State of Oregon.” The economic outcome Y_r might be whether the bill passed its originating chamber. The covariate W_r might be the party affiliation or roll-call voting score of the bill’s sponsor. ▲

Example: Financial news headlines Consider financial news headlines about publicly traded companies. Each text piece $r \in \mathcal{R}$ is a financial news headline such as “Bank of New York Mellon Q1 EPS \$0.94 Misses \$0.97 Estimate, Sales \$3.9B Misses \$4.01B Estimate.” The economic outcome Y_r might be the company’s realized return in some window after the headline’s publication date, while the covariate W_r could be the company’s past fundamentals. ▲

Each text piece r expresses some economic concept $V_r := f^*(r)$, where $f^*(\cdot)$ is the measurement procedure the researcher would use if time and resources were no constraint. Typically these measurements are what the researcher themselves or an appropriate domain expert would produce by carefully reading each text piece.² In defining the economic concept, researchers must answer: what exactly am I measuring from the text, and how would I label it if resources were no constraint? Moving forward, we assume a measurement procedure $f^*(\cdot)$ exists that the researcher would be satisfied to use *if* it could be scaled.

This creates a text processing problem: measuring the economic concept requires processing each text piece r , which may be prohibitively costly at scale. Absent a solution to this text processing problem, the researcher cannot collect V_r on all text pieces.

In settings like this, researchers would like use the collected text pieces to tackle two types of empirical analyses. The first is a *prediction problem*: predict the linked variable Y_r using the associated text piece r . For example, can we predict stock returns from news

²For example, [Ash and Hansen \(2023\)](#) write, “The most accurate approach to concept detection is perhaps direct human reading with appropriate domain expertise” (p. 672). [Hansen et al. \(2023\)](#) write, “The most precise way of classifying [text pieces] is arguably via direct human reading” (p. 6).

headlines? The second is an *estimation problem*: estimate some parameter that relates the economic concept V_r to the linked variables (Y_r, W_r) . If bill descriptions r express the policy topic V_r of the bill (e.g., whether it is related to foreign affairs or health policy), how do policy topics relate to the party affiliation of the bill’s sponsor W_r ?

LLMs are general-purpose and easy-to-use models for processing text, so researchers would like to use them in prediction and estimation problems. We next introduce LLMs into our framework.

2.2 Incorporating Large Language Models in Empirical Research

To capture how we often interact with LLMs as black boxes — generating responses from prompts without knowing their design nor training data — we define a *large language model* as any mapping from possible training datasets t to mappings between strings, where $\widehat{m}(\cdot; t): \Sigma^* \rightarrow \Sigma^*$ is its text generator when trained on dataset t and $\widehat{m}(\sigma; t)$ is the LLM’s response when prompted by string σ .

This definition has a key implication: the LLM is actually two distinct algorithms. First, a training algorithm that takes any dataset t and learns the mapping between strings. Second, the text generator $\widehat{m}(\cdot; t)$ is the output of the training algorithm and what users interact with.

Our analysis will not depend on how exactly these algorithms work. Since the state of the art is constantly evolving, we should expect the exact implementation of training algorithms and text generators to change. Furthermore, alternative LLMs may differ in their implementations. Studying LLMs at this level of abstraction will provide interpretable conditions for empirical research and ensures the durability of our analysis. Researchers will have conditions under which any model’s output can be incorporated into empirical research.

Our framework captures all key LLM design choices — some made by researchers, others by algorithm builders (often invisibly to researchers).

Interpreting the Text Generator The text generator $\widehat{m}(\cdot; t)$ captures all choices that influence how an LLM generates responses once trained. This includes prompt engineering strategies that materially affect the quality of responses (e.g., [Liu et al., 2023](#); [Wei et al., 2024](#); [White et al., 2023](#); [Chen et al., 2024](#)). Alternative prompt engineering strategies can be cast as alternative specifications of the text generator $\widehat{m}(\cdot; t)$.

The text generator captures other important choices governing generation. Parameters like temperature, top- p sampling, or top- k sampling control randomness in sampling from the LLM’s probability distribution over next tokens. Reasoning models employ test-time computation, generating intermediate “thought tokens” before producing final outputs to

enhance performance on complex tasks.³ By defining the text generator as a deterministic mapping from prompts to responses, our framework can be interpreted as focusing on the case in which these parameters are such that the LLM greedily generates its most likely token. Our results extend naturally to stochastic text generators as well, but at the expense of more cumbersome notation.⁴

Interpreting the Training Algorithm The training algorithm captures all aspects of design and training that produce the text generator. This includes architectural choices (e.g., parameter count, attention layers, context window), the pretraining objective (typically next-token prediction), and optimization details. LLMs undergo multiple training stages beyond pretraining. Instruction fine-tuning trains models to follow user instructions, and reinforcement learning from human feedback (RLHF) aligns output with human preferences. More recent developments include reinforcement learning from verifiable rewards (RLVR), where models are trained on tasks with objectively correct answers. Crucially, the training dataset t in our framework encompasses all strings used in pretraining *and* post-training stages.

So how do researchers use LLMs in empirical research? In prediction problems, researchers often prompt an LLM with each text piece r to generate a prediction $\hat{Y}_r = \widehat{m}(r; t)$. For instance, they might prompt a model to predict whether a stock return will be positive. In estimation problems, researchers prompt an LLM to generate labels $\hat{V}_r = \widehat{m}(r; t)$ for the economic concept. For example, they might prompt a model to classify a bill’s policy topic.

2.3 The Challenge of Evaluating Large Language Models

At some level, the viability of using LLMs for prediction and estimation problems hinges on the quality of LLM outputs: how large are possible errors $Y_r - \hat{Y}_r$ and $V_r - \hat{V}_r$ in any application? A natural starting point is to understand how computer science approaches this evaluation problem.

Computer scientists adapted methods from supervised learning. In supervised learning, the “common task framework” (Donoho, 2024) builds benchmark datasets like ImageNet for image classification and the Netflix Prize for recommender systems and evaluates competing algorithms on these benchmarks. Since LLMs aspire to be general-purpose technologies useful across all tasks, computer scientists have built increasingly diverse benchmarks.

³For many reasoning models, users cannot control generation parameters like temperature. Consequently, there is inherent randomness that cannot be eliminated.

⁴A practical challenge is ensuring LLM outputs are formatted appropriately for empirical analysis. Researchers often prompt LLMs to return structured outputs, but LLMs may not reliably comply. Our text generator abstraction takes further post-processing steps as given, and $\widehat{m}(\cdot; t)$ represents the resulting final output.

For example, the "Beyond-the-Imitation-Game benchmark" (BIG-bench) collects problems across 204 tasks (Srivastava et al., 2022), while the "Massive Multitask Language Understanding" (MMLU) benchmark spans 57 categories from logic to social sciences (Hendrycks et al., 2020). Other benchmarks assess specialized capabilities: SWE-bench for coding ability (Jimenez et al., 2024), GSM8K for mathematical reasoning (Cobbe et al., 2021), and standardized exams such as the SAT, GRE, and AP tests. Modern LLMs perform remarkably well on these benchmarks, and this impressive performance forms much of the quantitative basis for current enthusiasm.

Can economists use these benchmark evaluations to reason about how LLMs will perform on specific empirical tasks? The answer is more complicated than it might initially appear.

2.3.1 The Limits of Benchmarks

We do not intrinsically care about benchmark performance itself — after all, who deploys LLMs to solve SAT problems? We hope to *generalize* from benchmarks to new economically relevant tasks.

This generalization happens intuitively. We assume an LLM that aced AP chemistry must, like a person, handle many chemistry-related tasks. Evidence for "anthropomorphic generalization" suggests that people apply similar generalization heuristics to LLMs as they do to humans when predicting performance across tasks (Vafa, Rambachan and Mullainathan, 2024; Dreyfuss and Raux, 2024).

Yet this intuition is misleading. LLMs exhibit what Mancoridis et al. (2025) call "potemkin understanding" — performing well on benchmarks without grasping underlying concepts. This manifests as remarkable brittleness: performance is sensitive to seemingly minor details that would not affect human performance. Consider several examples from an accumulating body of evidence. While humans who master one math problem typically solve easier variations, LLMs have not readily generalized this way. An LLM that could reliably solve $(9/5)x + 32$ could not solve $(7/5)x + 31$ (McCoy et al., 2024). An LLM correctly defines an ABAB rhyming scheme but then generates poems violating its own definition (Mancoridis et al., 2025). An LLM trained on "A is B" will not know "B is A" (Berglund et al., 2023) and LLMs struggle with logical puzzle variations (Nezhurina et al., 2024). Brittleness extends to presentation: LLMs answer multiple-choice questions correctly but fail when answer order is permuted (Zong et al., 2024), succeed at programming with 0-based indexing but fail with 1-based indexing (Wu et al., 2024), and show inconsistent performance on similar spatial reasoning tasks (Mitchell, 2023).

LLM errors align poorly with human intuitions precisely because we expect them to understand and misunderstand as humans do. Dell’Acqua et al. (2023) aptly characterize

this as AI’s “jagged frontier” – a landscape where performance on similar tasks varies unpredictably. This jagged frontier captures why benchmark performance cannot be intuitively generalized to specific economic applications.

2.3.2 Do Large Language Models have World Models?

Perhaps this is too pessimistic, and this brittleness is superficial – noise around a deeper structural understanding embedded within LLMs. LLMs are trained on massive corpora containing rich information about reality, followed by extensive reinforcement learning. Perhaps LLMs have successfully learned “world models” – internal, structured representations of how the world works – that would enable reliable generalization despite occasional errors.

This hypothesis is an active research area focused on settings where researchers can both infer the LLM’s implicit world model and compare it to the truth (e.g., [Li et al., 2024](#); [Nanda, Lee and Wattenberg, 2023](#); [jylin04 et al., 2024](#); [Nikankin et al., 2025](#)). The evidence so far is at best mixed and at worst suggests LLMs may not learn generalizable world models. For example, [Vafa et al. \(2024\)](#) trained a generative sequence model on turn-by-turn driving data from 12.6 million NYC taxi rides. While it predicts the next turn between locations with high accuracy, the authors reverse-engineered the model’s implicit map of Manhattan’s street grid: the implicit map bears no resemblance to the actual grid. High predictive accuracy did not require learning the underlying spatial structure. [Vafa et al. \(2025\)](#) trained a generative sequence model on planetary orbits. Despite accurately predicting planetary movements, the model reveals no understanding of Newtonian physics. Rather than learning the gravitational law, it relies on task-specific heuristics that fail to generalize beyond its training distribution. Even advanced reasoning models exhibited similar failures.

Taken together, the evidence suggests that LLMs not only make errors, but these errors cannot be reliably predicted. They fail in ways misaligned with how humans generalize across tasks, nor can we assume these errors are noise around accurate world models. This creates our challenge. Since LLMs are impressive yet brittle tools, how should researchers incorporate their outputs into empirical research?

3 Prediction with Large Language Models

We begin with prediction problems – using text to predict economic outcomes. LLMs’ extensive training makes them natural candidates for this task, having already learned rich representations of language. Valid inference in prediction problems hinges on one condition: no leakage between the model’s training data and the researcher’s evaluation sample. While training leakage plagues benchmark evaluations in computer science, economists can manage it through careful model choice and research design.

3.1 The Researcher’s Prediction Problem

Suppose the researcher prompts an LLM to predict the linked variable from text pieces, $\hat{Y}_r = \widehat{m}(r; t)$.⁵ For some loss function $\ell(y, \tilde{y})$, the researcher calculates the sample average loss

$$\frac{1}{N} \sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r; t)), \quad (1)$$

where $N = \sum_r D_r$ is the number of text pieces collected. We would like to draw conclusions about the predictability of Y_r from the text piece r based on the LLM’s sample average loss. When is this valid?

Researchers face two sources of uncertainty about LLMs: What data was the LLM trained on? How does it generate output from text? We introduce two objects to formalize these uncertainties. The research context formalizes what we know (and do not know) about the LLM’s training data. The LLM’s guarantee summarizes high-level properties about how the model behaves without requiring exact knowledge of its internal workings. Together, these will allow us to derive interpretable conditions for using LLMs in prediction problems.

The *research context* $Q(\cdot) \in \mathcal{Q}$ is a joint distribution over the sampling indicators (D, T) . It encodes two distinct features. The sampling distribution over D defines the researcher’s out-of-sample prediction problem – the collection of text pieces over which they want to evaluate the LLM’s predictive performance. The sampling distribution over T captures the researcher’s uncertainty about the LLM’s training data. We make a technical assumption about the collection of research contexts $\mathcal{Q}$.

Assumption 1. Letting $t = (t_{\sigma_1}, \dots, t_{\sigma_{|\Sigma^*|}})'$ denote the sampling indicators summarizing the LLM’s realized training dataset, all research contexts $Q(\cdot) \in \mathcal{Q}$ satisfy: (i) For all values d , $Q(D = d, T = t) = \Pi_{\sigma \in \Sigma^*} Q(D_\sigma = d_\sigma, T_\sigma = t_\sigma)$; and (ii) $\mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r] = \mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r \mid T = t]$.

Assumption 1(i) states that sampling across strings is independent but not necessarily identically distributed. Assumption 1(ii) states that the researcher’s expected sample size does not depend on the LLM’s training corpus. Let $q_\sigma^{T|D}(t_\sigma) = Q(T_\sigma = t_\sigma \mid D_\sigma = 1)$ denote the conditional probability the string is sampled by the LLM’s training dataset given that it is sampled by the researcher. The marginal probabilities are $q_\sigma^T(t_\sigma) = Q(T_\sigma = t_\sigma)$ and $q_\sigma^D = Q(D_\sigma = 1)$.

Our goal is to assess the LLM’s predictive performance over the population of text pieces

⁵Researchers may use an LLM to construct embeddings for each text piece r , and the resulting embeddings may then be used as features by a supervised machine learning algorithm to predict Y_r . Our analysis equally applies to evaluating the performance of a prediction function using LLM embeddings as features.

defined by the research context $Q(\cdot)$:

$$\mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r; t)) \right]. \quad (2)$$

This summarizes whether the model’s predictive performance on the broader target population. As the number of economically relevant text pieces grows large (see Appendix C.1), we can characterize what the sample average loss converges to. Conditional on the LLM’s realized training dataset,

$$\frac{1}{N} \sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r; t)) - \frac{1}{\mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r \mid T = t]} \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r; t)) \mid T = t \right] \xrightarrow{p} 0. \quad (3)$$

Conditioning on the training dataset t treats the LLM $\widehat{m}(\cdot; t)$ as a fixed mapping, avoiding strong assumptions about how the LLM was trained (e.g., how would it behave over counterfactual training datasets?). The sample average loss recovers our target quantity in Equation (2) if $\mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r; t)) \mid T = t] = \mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r; t))]$.

Recall the second source of uncertainty: researchers do not know precisely how the LLM generates outputs from text. Yet researchers often observe other high-level information about the LLM’s behavior, such as performance on benchmarks (e.g., "achieves 88.7% on MMLU" or "scores 1520 on the SAT"). We formalize this through a *guarantee* $\mathcal{M}$, a collection of possible text generators that captures what the researcher knows about the LLM’s behavior. The researcher only knows that $\widehat{m}(\cdot; t) \in \mathcal{M}$.

Given a guarantee $\mathcal{M}$ and a research context $Q(\cdot)$, we define when this workflow is valid for prediction problems.

Definition 1 (Prediction problem). The LLM $\widehat{m}(\cdot; t)$ with guarantee $\mathcal{M}$ *generalizes* in research context $Q(\cdot)$ if, for all text generators satisfying the guarantee $\widehat{m}(\cdot) \in \mathcal{M}$,

$$\mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \ell(\widehat{m}(r), Y_r) \mid T = t \right] = \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \ell(\widehat{m}(r), Y_r) \right].$$

This definition formalizes an out-of-sample prediction goal. The LLM generalizes if its sample average loss reflects its predictive performance on the target population. Under Definition 1, the researcher’s workflow in the prediction problem is justified for *any* LLM satisfying the guarantee $\mathcal{M}$. The researcher can draw conclusions based on the sample average loss knowing only the guarantee $\mathcal{M}$ is satisfied.

Example: Congressional legislation Consider researchers predicting whether a bill passes either house of Congress Y_r using only its text description r . The LLM generates

predictions $\widehat{m}(r; t)$ using a specific prompting strategy. Each researcher calculates the sample average loss of the LLM’s predictions on their own collected sample of Congressional bills. Different researchers face different prediction problems, formalized by alternative research contexts $Q(\cdot)$. One researcher might assess whether the LLM can predict outcomes for future legislation by sampling bills after a cutoff date, such as all bills from 2025 onward. Another might evaluate whether the LLM’s predictions generalize to novel policy domains, sampling bills on as cryptocurrency regulation or artificial intelligence policy. ▲

Example: Financial news headlines Consider researchers predicting a company’s realized returns Y_r from a financial news headline r . The LLM generates predictions $\widehat{m}(r; t)$ using a specific prompting strategy. Each researcher calculates the sample average loss on their collected headlines. Different researchers again face different out-of-sample prediction problems, formalized by alternative research contexts $Q(\cdot)$. One researcher might assess whether the LLM predicts returns during future market conditions, sampling headlines from 2025 onward. Another might evaluate whether predictions generalize across company types, focusing on small-cap technology firms or international equities. ▲

3.2 Training Leakage as a Threat to Prediction

Using LLMs in prediction problems requires one condition: no training leakage between the LLM’s training data and the researcher’s evaluation sample.

Lemma 1. *Under Assumption 1, for any research context $Q(\cdot) \in \mathcal{Q}$ and text generator $\widehat{m}(\cdot)$,*

$$\mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r)) \right] = \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \ell(Y_r, \widehat{m}(r)) \mid T = t \right] - \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \left(\frac{q_r^{T|D}(t_r)}{q_r^T(t_r)} - 1 \right) \ell(Y_r, \widehat{m}(r)) \right].$$

Proposition 1. *The LLM $\widehat{m}(\cdot; t)$ generalizes for research context $Q(\cdot) \in \mathcal{Q}$ if and only if it satisfies the guarantee $\mathcal{M}(Q)$ for*

$$\mathcal{M}(Q) = \left\{ \widehat{m}(\cdot): -\mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \left(\frac{q_r^{T|D}(t_r)}{q_r^T(t_r)} - 1 \right) \ell(Y_r, \widehat{m}(r)) \right] = 0 \right\}. \quad (4)$$

The no-training leakage condition (Equation 4) captures the extent to which overlap between the LLM’s training data and the researcher’s sample covaries with the LLM’s predictive performance. It has an intuitive interpretation as omitted variable bias. The term $\frac{q_r^{T|D}(t_r)}{q_r^T(t_r)} - 1$ measures how learning that the researcher sampled text piece r updates beliefs about whether r appeared in the LLM’s training data. When this correlation is positive – text pieces in the researcher’s sample are more likely to have been in the training data – and the LLM performs well on such pieces, the overall bias term is positive and the sample average loss overstates

true predictive performance. Uncertainty over what entered into the LLM’s training data acts like an omitted variables bias in the prediction problem.

This bias arises from both the researcher’s choice of out-of-sample prediction and our beliefs about what entered into the LLM’s training data. Section 3.3 discusses how researchers can prevent training leakage through careful choices of model and prediction exercise.

3.2.1 Some Evidence of Training Leakage

Training leakage is a well-documented problem in computer science. LLMs’ training datasets frequently contain examples from popular benchmark evaluations (Sainz et al., 2023; Golchin and Surdeanu, 2024; LM Contamination Index, 2024). This has generated skepticism about evaluating LLMs on any publicly available data (e.g., Ravaut et al., 2024). But this evidence focuses on CS applications. Might economic prediction problems face similar risks?

Assessing Training Leakage in Congressional Legislation We assess training leakage in an empirical setting relevant for economists studying politics and political economy: congressional legislation. The Congressional Bills Project (Wilkerson et al., 2023; Adler and Wilkerson, 2020) contains descriptions r for over 400,000 bills proposed in Congress, along with whether each passed the House or Senate Y_r . We sample 10,000 bills introduced from 1973 to 2016 and test whether passage can be predicted from text descriptions alone — a potentially challenging task given the strategic dynamics of congressional voting. Among these bills, only 7.4% pass the House and 6.0% pass the Senate.

We generate predictions $\hat{Y}_r = \widehat{m}(r; t)$ based on each bill’s description r by prompting GPT-4o (see Appendix Figure A13 for the specific prompt). GPT-4o correctly predicts the bill’s outcome 91.2% of the time in the House and 92.5% of the time in the Senate (left panel of Appendix Table A1). What drives the model’s ability to accurately predict whether a Congressional bill will pass the House or the Senate based only on its description? The answer is that the text of congressional legislation is likely included in its training dataset.

To evaluate training leakage, we prompt the model to complete each bill’s text description based on only the first half of its text, following research in computer science such as Golchin and Surdeanu (2024) (see Appendix Figure A14 for the specific prompt). If a model reproduces the text exactly, it has likely seen it during training. This is not a necessary condition for training leakage; models may benefit from exposure to a text piece without memorization. Perfect reproduction nonetheless offers compelling evidence.

On 344 bills, GPT-4o completes the bill’s text description *exactly* as it is written, indicating that not only was GPT-4o likely trained on these text pieces but it appears to have memorized them. Appendix Figure A1 provides two examples of successful completions. On

the other bills, GPT-4o’s completed bill descriptions are close to the original bill descriptions. Word embeddings of GPT-4o’s descriptions are far closer to those of the originals than the average distance between the word embeddings of two randomly selected bills (left panel of Appendix Table A2).

One might wonder whether this training leakage could be addressed through prompt engineering — for example, by commanding the models to not pay attention to any information past a certain date. Applying this prompt engineering strategy to our sample of Congressional bills (see Appendix Figure A13 and A14 for associated prompts), we still find substantial evidence of training leakage. Even when explicitly told to not consider any information past the bill’s introduction date in Congress, GPT-4o can still accurately predict its outcome in the House and the Senate based on these small snippets of text (right panel of Appendix Table A1). GPT-4o still exactly completes nearly the same number of bill descriptions as without the prompt engineering (330 versus 344), and the word embeddings of its completed descriptions remain quite close on average to those of the originals (right panel of Appendix Table A2).

Assessing Training Leakage in Financial News Headlines We consider another domain relevant for economists: financial markets. Existing work (e.g., Glasserman and Lin, 2023; Lopez-Lira and Tang, 2024) found that LLMs predict stock returns accurately from news headlines. We test whether this reflects training leakage using publicly available headline data (Aenlle, 2020) covering nearly 4 million headlines for 6,000 publicly traded companies from 2009-2020.

We sampled 10,000 financial news headlines from 2019, and we prompt GPT-4o to complete each financial news headline based on only 50% of its text (see Appendix Figure A15 for the specific prompt). GPT-4o reproduces 60 financial news headlines exactly as they were written in the publicly available dataset, indicating that GPT-4o was likely trained on these headlines and memorized them. Appendix Figure A2 provides two examples of successfully completions. On all other headlines, word embeddings of the model’s completions are close to those of the original headlines (left panel of Appendix Table A3).

We again explore whether explicitly incorporating date restrictions into the LLM prompt moderate this evidence of training leakage (see Appendix Figure A15 for the associated prompts).⁶ Surprisingly, this appears to make the problem *worse*; GPT-4o now reproduces 73 headlines exactly, and word embeddings of GPT-4o’s completions with the date restriction are still on average close to those of the original headlines.

⁶See also Wongchamcharoen and Glasserman (2025). Another prompting strategy – entity masking – aims to prevent memorized information by masking identifiers in prompts (Glasserman and Lin, 2023; Sarkar and Vafa, 2024; Engelberg et al., 2025). However, implementation details matter, and its wider applicability is unclear. In our congressional legislation example, it is not obvious what should be masked.

Sarkar and Vafa (2024) provide additional evidence of training leakage in finance: prompting Llama 2 to predict risks from September-November 2019 earnings calls, the LLM mentions Covid-19 in over 25% of cases—a form of "look-ahead bias" from training on future information. Combined with our findings and other evidence (Glasserman and Lin, 2023; Lopez-Lira, Tang and Zhu, 2025), this indicates substantial risk of training leakage in finance applications.

3.3 Practical Guidance for Prediction Problems

The no-training-leakage condition (Equation 4) may seem abstract, but it provides concrete guidance for empirical practice. Researchers must consider what population they are predicting on, how their evaluation sample relates to it, and how their evaluation sample relates to the LLM’s training corpus.

To make this concrete, recall that the bias term in Proposition 1 depends on $\frac{q^{T|D}(t_r)}{q^T(t_r)} - 1$ – how much does learning that the researcher sampled text piece r update our beliefs about whether r was in the training data. No training leakage requires this term to be zero, or at least uncorrelated with the LLM’s performance. Different prediction problems and model choices achieve this in different ways. We illustrate this by considering the threat of training leakage and how to manage it across several examples.

Example: Lookahead Bias and Time-Stamped Models Consider a researcher who wants to predict stock returns from future financial news headlines. If the researcher evaluates on recent headlines from 2024 using an LLM trained on data through 2025, the leakage term is almost certainly positive: both the researcher and the LLM builders had access to 2024 headlines. Any predictive success may reflect training leakage.

The solution in this case is to use open-source LLMs with fixed published weights, such as the Llama family (Touvron et al., 2023; Dubey et al., 2024) and others (e.g., BLOOM, OLMo, etc.), or time-stamped training data (e.g., Sarkar, 2024; He et al., 2025). If the researcher uses a model with weights published on date τ and constructs an evaluation sample using documents published after τ , then $q_r^T(t_r) = 0$ mechanically since they could not have been in the training data. This eliminates training leakage by design.

Closed models like the GPT family from OpenAI or the Claude family from Anthropic do not permit this solution. Their training data is undisclosed, and they may be continuously fine-tuned. Researchers in natural language processing now regularly caution against sending test data to the APIs or chat interfaces of closed LLMs since these data may be used in further fine-tuning or the development of new models (Jacovi et al., 2023).⁷ Most

⁷Balloccu et al. (2024) estimates that 263 benchmarks may have been inadvertently leaked to OpenAI through use of the chat interface and API, and Cheng et al. (2024) questions the validity of publicly stated

worryingly, [Barrie, Palmer and Spirling \(2024\)](#) illustrates that the results of submitting the same prompt to GPT-4o yields results that change month-to-month, despite there being no publicly announced changes to the underlying model. ▲

Example: Prediction on Confidential Documents Suppose a researcher wants to predict, for example, whether administrative case notes are predictive of some decision, like pretrial release. In examples like this, the target population consists of confidential documents that were never publicly released. Since these documents never entered any public corpus, we can again credibly argue that $q_r^T(t_r) = 0$ for all documents since they were never available for training. This reasoning hinges on credibly claiming the documents’ provenance. ▲

Example: Random Sampling from a Known Corpus Consider a researcher with a complete corpus of economics papers published between 2000 and 2020 who wants to predict citation counts from abstracts. She draws a random sample to form her evaluation set. In this case, the researcher’s sampling process is independent of the LLM’s training data: her random number generator knows nothing about which papers OpenAI included in their training data. This implies $q_r^{T|D}(t_r) = q_r^T(t_r)$ – learning that a paper was randomly selected provides no information about whether it was trained on — so the leakage term equals zero.

The logic is familiar from causal inference: randomization eliminates omitted variable bias by making treatment assignment independent of potential outcomes. Here, random sampling eliminates training leakage by making the researcher’s evaluation sample independent of the LLM’s training data. This can be applied whenever the researcher draws a genuinely random sample from a well-defined corpus before using an LLM. The converse is also important: just as non-random treatment assignment resurrects possible confounding in causal inference, non-random sampling may resurrect training leakage. ▲

Example: Prediction on the Complete Population As a final example, suppose the researcher collected all Congressional bill descriptions from 1973–2016 and wants to understand what textual features predict passage. The researcher seeks to descriptively characterize patterns in this specific historical corpus, not to make predictions on some population of legislation. When the researcher observes the entire population, the prediction problem changes character. There is no sampling, and hence no inference to a broader population. In this case, $q_r^D = 1$ and the leakage condition is satisfied trivially: we are not asking whether performance generalizes beyond what we observe. The complete-population case licenses only descriptive conclusions about the collected corpus. ▲

“knowledge cutoffs” in closed LLMs.

More broadly, these examples illustrate a general principle: no training leakage is ultimately about *both* research design and model properties. Researchers must be explicit about their target population, their sampling procedure, and how that procedure relates to possible training corpora. When in doubt, the safest approach combines open-source models with published weights or documented training data and evaluation samples constructed to mechanically exclude the model’s training corpus.

4 Estimation with Large Language Models

In estimation problems, the researcher measures economic concepts V_r expressed in text pieces r to estimate downstream parameters. The measurement $f^*(\cdot)$ is costly to scale across thousands or millions of text pieces. This is where LLMs enter as potential substitutes.

Using LLM outputs in plug-in estimation requires a strong assumption: the LLM must reproduce the existing measurement everywhere. As Section 2.3 discussed, LLM performance varies across similar tasks, and benchmark evaluations provide little guidance on new applications. We demonstrate this empirically: seemingly minor choices – which LLM, which prompt – substantially affect downstream parameter estimates in applications to finance and political economy. These choices change the magnitude, significance, and even sign of estimated parameters. The solution: collect a small validation sample and use it to debias the LLM’s outputs.

4.1 The Researcher’s Estimation Problem

The researcher specifies a parameter $\theta \in \Theta$ and a moment condition $g(\cdot)$ that identifies this parameter. If she collected the economic concept V_r on each text piece r , she would calculate

$$\hat{\theta}^* = \arg \min_{\theta \in \Theta} \frac{1}{N} \sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta). \quad (5)$$

For example, letting $g(V_r, W_r; \theta) = (V_r - W_r' \theta)^2$, the researcher studies how V_r relates to the linked variables W_r , and $\hat{\theta}^*$ is the sample regression coefficient. Due to the text processing problem, however, the researcher cannot calculate $\hat{\theta}^*$ directly.

The researcher prompts an LLM to measure the economic concept on each text piece, $\hat{V}_r := \widehat{m}(r; t)$, and plugs in the LLM’s labels

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \frac{1}{N} \sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r; t), W_r; \theta). \quad (6)$$

When can we draw conclusions about $\hat{\theta}^*$ based on $\hat{\theta}$?

We associate the researcher with a research context $Q(\cdot) \in \mathcal{Q}$ satisfying Assumption 1.

We assume that the researcher could study any moment condition $g(\cdot) \in \mathcal{G}$ satisfying the following assumption.

Assumption 2. For all $g(\cdot) \in \mathcal{G}$, $g(\cdot)$ is differentiable and there exists some $\bar{G} > 0$ such that $\left| \frac{\partial g(v, W_r; \theta)}{\partial \theta} \right| \leq \bar{G}$ for all $r \in \mathcal{R}$, $\theta \in \Theta$, and values of the economic concept v .

As the number of economically relevant text pieces grows large (see Appendix C.1), we can characterize the behavior of both estimators. The target moment condition converges to, at any parameter value $\theta \in \Theta$,

$$\frac{1}{N} \sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) - \frac{1}{\mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r]} \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) \right] \xrightarrow{p} 0. \quad (7)$$

Conditional on the LLM’s realized training dataset, the plug-in moment condition converges to, at any parameter value $\theta \in \Theta$,

$$\frac{1}{N} \sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) - \frac{1}{\mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r \mid T = t]} \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r; t), W_r; \theta) \mid T = t \right] \xrightarrow{p} 0. \quad (8)$$

As in the prediction problem, conditioning on the training dataset simplifies analysis by treating the LLM as a fixed mapping.

Given an LLM with guarantee $\mathcal{M}$, the researcher would like to recover the moment condition defined using the economic concept.

Definition 2. The LLM $\widehat{m}(\cdot; t)$ with guarantee $\mathcal{M}$ *automates* the existing measurement $f^*(\cdot)$ for the moment condition $g(\cdot)$ in research context $Q(\cdot)$ if, for all models satisfying the guarantee $\widehat{m}(\cdot) \in \mathcal{M}$ and all $\theta \in \Theta$,

$$\mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r), W_r; \theta) \mid T = t \right] = \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) \right].$$

The LLM $\widehat{m}(\cdot; t)$ with guarantee $\mathcal{M}$ is a *general-purpose technology for estimation* if it automates the researcher’s measurement process for all $g(\cdot) \in \mathcal{G}$ and $Q(\cdot) \in \mathcal{Q}$.

The excitement around LLMs stems from their potential as “general-purpose technologies” – tools deployable across diverse applications without task-specific engineering (e.g., Eloundou et al., 2024). For estimation problems, this would mean reliably substituting for existing measurements across different economic concepts and research contexts. Definition 2 formalizes what this requires: the LLM must produce moment conditions matching the existing measurement procedure, regardless of which moment condition or population the researcher studies. If an LLM satisfies this property, researchers could confidently use its outputs for plug-in estimation knowing only the guarantee $\mathcal{M}$.

4.2 Measurement Error as a Threat to Estimation

We clarify what guarantee $\mathcal{M}$ is necessary and sufficient for an LLM to automate the existing measurement. We decompose the difference between the plug-in moment condition and the target moment condition into two terms.

Lemma 2. *Under Assumption 1, for any research context $Q(\cdot) \in \mathcal{Q}$, moment condition $g(\cdot) \in \mathcal{G}$ and text generator $\widehat{m}(\cdot)$, $\mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r), W_r; \theta) \mid T = t] - \mathbb{E}_Q[\sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta)]$ equals*

$$\left(\mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r), W_r; \theta) \right] - \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) \right] \right) + \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r \left(\frac{q_r^{T|D}(t_r)}{q_r^T(t_r)} - 1 \right) g(\widehat{m}(r), W_r; \theta) \right]. \quad (9)$$

The second term captures training leakage – potential overlap between the LLM’s training dataset and the researcher’s evaluation sample. As discussed in Section 3.3, training leakage can be controlled through appropriate choice of model and research context. For example, if the researcher collects all text pieces in their research context or randomly samples text pieces, then training leakage is mechanically zero.

We therefore focus on the first term, which captures possible LLM errors. Intuitively, LLM errors $\Delta_r = \widehat{m}(r; t) - V_r$ can bias parameter estimates if they correlate with the economic variables W_r . A natural question arises: what if we know the LLM is “pretty good” – say, within some δ of the true measurement everywhere? This is what we might be tempted to conclude from impressive benchmark evaluations and demonstrations of LLM capabilities. More precisely, suppose the LLM satisfies the guarantee $\mathcal{M}(Q, \delta)$ — the collection of text generators satisfying $\|\widehat{m}(\cdot) - f^*(\cdot)\|_{\infty, Q} = \max_{r \in \mathcal{R}: q_r^D > 0} |\widehat{m}(r; t) - f^*(r)| \leq \delta$. Does this ensure valid plug-in estimation?

Unfortunately, knowing the guarantee $\mathcal{M}(Q, \delta)$ does not tell us about the exact pattern of errors — whether they relate to economic variables in ways that bias our estimates. We say that a moment condition $g(\cdot)$ is *sensitive* to the economic concept V_r in research context $Q(\cdot)$ if $q_r^D > 0$ and there exists some $\underline{G} > 0$ such that $|\frac{\partial g(v, W_r; \theta)}{\partial v}| \geq \underline{G}$ for all v, θ . Let $\mathcal{R}(g, Q)$ denote the collection of sensitive text pieces.

Lemma 3. *Consider any moment condition $g(\cdot) \in \mathcal{G}$ in research context $Q(\cdot) \in \mathcal{Q}$. Then, for all $\theta \in \Theta$ and $\widehat{m}(\cdot) \in \mathcal{M}(Q, \delta)$ satisfying no training leakage,*

$$\left| \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r), W_r; \theta) \mid T = t \right] - \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) \right] \right| \leq \overline{G} \delta. \quad (10)$$

But, for all $\theta \in \Theta$, there exists $\widehat{m}(\cdot) \in \mathcal{M}(Q, \delta)$ that satisfies no training leakage such that,

for $\delta(r)$ defined in the proof,

$$\left| \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(\widehat{m}(r), W_r; \theta) \mid T = t \right] - \mathbb{E}_Q \left[\sum_{r \in \mathcal{R}} D_r g(V_r, W_r; \theta) \right] \right| \geq \underline{G} \left(\sum_{r \in \mathcal{R}(g, Q)} |\delta(r)| q_r^D \right). \quad (11)$$

Lemma 3(i) shows guarantee $\mathcal{M}(Q, \delta)$ bounds the LLM's error for the moment condition. But this is not enough for the LLM to automate the existing measurement. Lemma 3(ii) shows text generators satisfying the guarantee $\mathcal{M}(Q, \delta)$ can still produce meaningful estimation errors. We cannot rule out that the LLM's errors correlate with economic variables in ways that bias estimates. This makes measurement error pernicious.

This leads to our main characterization. Researchers can safely ignore the details of the LLM's design no matter the research context studied and economic question being asked if and only if its labels reproduce the existing measurement process everywhere. Anything less cannot guarantee valid plug-in estimation across all possible applications.

Proposition 2. *Suppose the LLM $\widehat{m}(\cdot; t)$ satisfies no training leakage in all research contexts $Q(\cdot) \in \mathcal{Q}$ and moment conditions $g(\cdot) \in \mathcal{G}$. Provided there exists some $g(\cdot) \in \mathcal{G}$ that is sensitive to the economic concept for any $r \in \mathcal{R}$, then the language model is a general-purpose technology for estimation if and only if $\widehat{m}(\cdot; t)$ satisfies the guarantee $\mathcal{M}(Q, 0)$ for all research contexts $Q(\cdot)$.*

4.2.1 Some Evidence of Measurement Error

Plugging in LLM outputs requires that the model reproduces the target measurement process $f^*(\cdot)$. While their impressive performance on some tasks makes this assumption appealing, Section 2.3 showed such intuitions about LLMs are unreliable. No LLM achieves perfect accuracy on benchmark evaluations, and growing evidence documents their surprising errors. Does this matter for economics research? We next show that LLM errors substantially affect downstream parameter estimates in two empirical examples.

Linear Regression with LLMs Consider a researcher relating the economic concept V_r and linked variables W_r through linear regression. The researcher may use the LLM's labels as the dependent variable:

$$V_r = W_r' \beta^* + \epsilon_r, \text{ and } \widehat{m}(r; t) = W_r' \beta + \tilde{\epsilon}_r. \quad (12)$$

or the independent variable:

$$W_r = V_r' \alpha^* + \nu_r, \text{ and } W_r = \widehat{m}(r; t)' \alpha + \tilde{\nu}_r. \quad (13)$$